



University of Cabuyao
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COLLEGE OF COMPUTING STUDIES

**AUTOMATED REAL-TIME ACADEMIC DOCUMENT VERIFICATION
USING OCR-BASED TEXT EXTRACTION AND RULE-BASED
VALIDATION**

A Thesis Proposal

Submitted to the Faculty of
The College of Computing Studies
PAMANTASAN NG CABUYAO
City of Cabuyao, Laguna

In Partial Fulfillment
of the Requirements for the Degree:
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

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May 2026



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CHAPTER I

THE PROBLEM AND ITS SETTING

This is a research on the design of an Automated Real-Time Academic Document Verification System via OCR-Based Text Extraction, Rule-Based Validation uses tools which are used to read text in a scanned document. It examines and checks the text if everything is correct. The system verifies the absence of information, incorrect format and incomplete information right after it receives the documents. The Automated Real Time Academic Document verification system is highly helpful as it is able.

Introduction

Learning institutions are actually in need of documents these days due to the digitalizing of education. We must ensure that such documents as transcripts and enrollment certificates are accurate and complete. This is in order to be guided by the organization rules. Many registrar offices in the Philippines continue to do it manually and this process is time consuming and could involve errors. One of the studies is on the capability of OCR text and image extraction in automating processes of data entry to save time and minimize errors that may be caused by manually entering data. [49]

This is an issue, to remedy this, we are discussing a system that can check documents automatically in real time. The Automated Real-Time Academic Document Verification System is what we are proposing. This



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system involves computer techniques. It takes advantage of a process known as Optical Character Recognition or OCR in short to retrieve scanned document text.. It uses the Tesseract engine to do this. The Automated Real-Time Academic Document Verification System is an aid since it may simplify and streamline the work of the registrar offices in the Philippines. The Automated Real-Time Academic Document Verification System may ensure that academic documents are right and observe the regulations.[1],[2],[3],[4].

Post processing of the image, like using Gaussian smoothing, binarization, and contrast enhancement (e.g., CLAHE) are used to enhance the accuracy of OCR [5],[6],[7],[8],[9]. Along with that, there is rule-based validation based on IF-THEN logic that allows making sure that extracted data meets with the institutional standards and demanded formats [14],[15],[16],[17].

With the help of these algorithms, the offered system helps to achieve accuracy, efficiency, as well as interpretability by ensuring all the limitations that are the subject of analysis in research are addressed but modifying it to the requirements of the particular issue of academic document verification in schools of higher education.

The system uses multimodal mechanisms other than text recognition as an additional verification method. With the help of OCR and Natural Language Processing (NLP), it is also possible to classify documents, and Convolutional Neural Networks (CNNs) are employed to identify visual attributes, including logos and seals [3], [11], [37], [38]. Use of deep learning is also used in signature verification to differentiate authentic and counterfeit signatures [40]. In addition, the verification and classification



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algorithms, including the Gradient Boosting Decision Trees (GBDT), aid in structural conformity and enhanced accuracy of validations [42].

In spite of these developments, issues like image quality, computation resource demands, and organizational flexibility still exist [11],[12],[13],[14],[15]. Even the existing systems, as a rule, pay attention to the general or government-issued documents, not focusing on academic records. As such, a system that is particularly meant to handle the operations of the registrar in Philippine institutions is required.

To address these problems, we are proposing an Automated Real-Time Academic Document Verification System. This system will ensure that the academic documents are accurate and that the registrar office can carry out its operations effectively and freely. The OCR is used in the system to extract the text in scanned documents, and this is of great assistance. It also compares the information it receives to the school rules to ensure that such details as name and date are accurate. Automated Real-Time Academic Document Verification System can be applied to schools since it has a great number of various methods of checking documents and it can discuss how it works. This explains why it is a good fit in the requirements of the schools in the Philippines and the Automated Real-Time Academic Document Verification System.



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Statement of the Problem

The situation with academic document verification in the registrar offices remains largely the same, as manual processes cannot be effectively used as far as accuracy, timeliness, and compliance are concerned. These are all traditional processes that are labor intensive, time consuming and subject to human error, leading to delays, inconsistencies and inefficiencies in the provision of academic services. Since schools aim to maintain credibility and offer hardy assistance to students and teachers, the continuation of these archaic measures brings major issues to the stability of flowlined academic functioning.

To solve the problems, the research is expected to provide the answers to the following questions:

1. How can the OCR based text extraction be successfully adopted to extract and extract the relevant textual information in relation to scanned academic papers?
2. How is rule-based validation to be designed and applied in:
 - 2.1 Detect missing part;
 - 2.2 Unfilled Document; and
 - 2.3 non-compliant areas according to institutional policies?



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3. How can an automated real-time academic document verification system be developed to support the workflow of the academic environment?

4. What is the level of accuracy of the proposed system on checking academic documentation over traditional manual checking methods, especially with algorithms:

Convolutional neural networks (CNNs);

4.2 Long Short- Term Memory (LSTM) models;

4.3 hybrid methods of OCR and validation based on transformers?

The proposed system performs well with regard to:

5.1 Processing time; and

5.2 Performance efficiency?

6. How acceptable are the suggested system among staff in the registrar in terms of usability, reliability, and usefulness?

7. How have IT experts reviewed the technical feasibility, security and architectural integrity of the system?



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Scope and Limitation

The purpose of this project is to build and test an automated real-time system of academic document verification which relies on OCR-based text extraction and a rule-based verification. The main users of the system are the registrar members of Pamantasan ng Cabuyao. The study includes only a small number of academic documents, including transcripts, certificates, and enrollment-related paperwork which are regularly processed by the Registrar of the Office. The given system will be introduced and tested in the institution in the 2025-2026 school year.

The functionality of the system is created to automate the verification process. It scans printed scholarly materials with a technology known as Optical Character Recognition (OCR) to transform text found in the form of images into machine-readable text. This will allow the person working in the registering department to organize the documentation without retyping the details. Second, the system executes the validation of rules; this entails the enforcement of institutional policies through completeness, accuracy, and compliance checking. As an example, it is capable of automatically notifying the absence of student names, wrong course codes, or wrong dates. Third, the system produces real-time feedback reports, which gives the staff at the registers real-time notifications, that is, those reports that indicate any error or inconsistency. This feedback will ensure problems are rectified early enough, decreasing delays in academic services.

The system features make it have preprocessing modules, multimodal verification, and explainable decision making. Before OCR extraction,



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pre-processing methods like CLAHE, Gaussian smoothing, and Laplace sharpening are use to improve the quality of the image to make it easier to recognize the objects. Multimodal verification goes beyond recognition of text into the recognition of logs, seal recognition and signature recognition acts that fortify authenticity tests. These characteristics are based on image-processing algorithms and deep-learning models to detect elements of institutions and to separate authentic signatures and forgeries. Also, the system combines explainable AI systems which render available validation decisions to registrar, thus promoting reliability and usability.

The performance of the system is based on its algorithms. Engines like Tesseract, and EasyOCR are in use and EasyOCR uses Convolutional Neural Networks (CNNs) network and Long Short-Term Memory (LSTM) networks to process noisy and skewed images as well as handwritten texts. Combination of classical OCR with transformer-based approaches is also viewed as a way of helping to recognize complex academic records. Deterministic if-then logic is used to implement rule based validation, and guarantee the rules of an institution. In multimodal verification logo and seal detection is done using CNN-based algorithms, whereas signature verification is done using deep convolutional neural networks. The format verification uses the gradient boosting decision trees to verify the compliance with the institutional design standards. These algorithms combined make the system have a balance between accuracy, efficiency and transparency.

Limitations of the study can influence the extraction of information on texts. Document quality of scanned documents, change in format of documents, and accuracy of OCR are still among the critical factors. The system cannot be able to treat non-standard layouts, badly torn papers and



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handwritten text reliably. In addition, the validation rules are also limited to the existing policy of the Pamantasan ng Cabuyao and therefore may not readily apply to other institutions. Deep learning algorithms also have a high computational requirement, and institutions with a less developed infrastructure might not be able to cope with real-time processing.

Significance of the Study

This study provides both **academic and practical contributions** to the field of computer science, particularly in **automated document processing, OCR, and rule-based validation systems**. It benefits several stakeholders at different levels, from institutions down to individual users.

Macro level (Institutions and Academic Community): The educational institution would benefit from the proposed system in terms of increased efficiency, accuracy, and reliability in dealing with academic documents. By using the proposed system, the institution would be able to reduce waiting times, minimize human error, and increase their credibility in providing academic services to their students. Moreover, the academic community would benefit from the proposed system, as it would serve as a model of an academic system that utilizes an algorithm to solve real-world problems in the academic community, which would contribute to the digital transformation of the academic community.



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Meso level (Registrar Office and Academic Staff): The registrar staff of Pamantasan ng Cabuyao, as the primary users, directly benefit from the automation of document verification tasks. The system supports their existing workflow by reducing repetitive manual work, decreasing verification errors, and improving processing speed. It also provides real-time validation results, allowing staff to make faster and more accurate decisions. As a result, employees can focus on more complex administrative responsibilities rather than routine document checking.

Micro level (Students): Students are also direct beneficiaries of the system. They experience faster processing of academic documents such as transcripts, certificates, and enrollment forms. This leads to shorter waiting times, smoother transactions, and fewer delays in academic processes such as enrollment, graduation, and certification requests.

Researchers (Present and Future): This study significantly benefits **researchers**, especially in the field of computer science. It provides a foundation for exploring **intelligent document processing, OCR optimization, and rule-based validation systems**. The current researchers contribute by designing and implementing a system that integrates these computational techniques in an academic setting.

Future researchers- can use this study as a reference to develop more advanced document verification systems. They may enhance the system by integrating **machine learning algorithms, artificial intelligence models, and explainable AI techniques**. Additionally, future studies may improve OCR accuracy, expand validation rules, or adapt the system to handle different document formats and institutional requirements.



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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

In many Philippine universities, the old way of verifying documents tends to be slow, frankly, and vulnerable to mistakes. People checking piles of papers by hand can easily miss details or make errors. To address this, schools have started turning to automated systems that rely on OCR technology and rule-based checks, making sure that information is captured with precision. On the top of that, artificial intelligence and deep learning models are now being used to sort data, spot inconsistencies, and even flag suspicious signatures or forged records. The result is a streamlined process that doesn't just save time it also builds trust, ensuring that academic records remain secure, authentic, and reliable.

One of the most crucial responsibilities in any university is making sure academic papers are properly checked. Documents like transcripts, enrollment certificates, and diplomas aren't just pieces of paper; they determine whether a student can move forward, graduate, or even secure future career opportunities. Because of this, institutions place enormous value on their authenticity and accuracy. Every detail has to be complete, consistent, and aligned with the official standards. Without that guarantee, the credibility of the educational system itself and the legitimacy of the university would be at risk.



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For years, many universities and colleges in the Philippines, particularly registrar offices have relied on semi-manual methods to verify academic documents. Clerical staff typically go through each record by hand, checking fields against institutional policies and visually confirming whether the papers look legitimate. While this approach does get the job done, it's undeniably heavy on labor and time. The process often drags on, leaving room for human error and inconsistencies. As a result, delays pile up, irregularities slip through, inefficiencies affect not only the students waiting for their records but also the institutions trying to uphold their standards.

With the rapid digital transformation sweeping through education, universities now have more effective ways to address long-standing verification issues. Optical Character Recognition(OCR) has emerged as a powerful tool to extracting text from scanned or photographed academic documents [1], [2], [3], [4], [10], [11], [12], [22], [23], [24], [25]. To strengthen this process, rule-based validation techniques are applied, ensuring that the extracted information compiles with the institutional standards and logical requirements[14], [15], [16], [17], [26]. When these two technologies work hand in hand, they create a strong foundation for automating document verification making the process faster, more reliable, and far less prone to error.



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Recent research shows that verification today goes far beyond simply recognizing text. Modern systems integrate feature extraction and classification, combining OCR with Natural Language Processing(NLP) to categorize certificates effectively[36]. Institutional logos and seals have also become critical markers of authenticity, with specialized models designed for logo detection and identification [37], [38], [39]. Signature verification has advanced as well, with deep learning algorithms distinguishing genuine signatures from fraudulent ones [40]. Even formatting is scrutinized, ensuring that documents adhere to institutional design standards [42]. At the highest level, legal document authentication systems weave together OCR, NLP, and convolutional neural networks(CNNs) to uncover forgeries across textual, visual, and structural layers [41].

This chapter takes a closer look at the body of research, related studies, and the conceptual groundwork surrounding automated verification of academic documents. It not only reviews existing literature but also lays out the theoretical foundation that supports the proposed system. In addition, the chapter introduces the conceptual framework, showing how different components, technologies, methods, and validation strategies fit together to form a coherent approach. In essence, it bridges prior scholarship with the design of the system being suggested, ensuring that the project rests on solid academic and practical principles.



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Conceptual Literature

The proposed system incorporates the core features typically found in automated document verification platforms text extraction, data validation, document classification, and the generation of results. Together, these functions streamline the handling of academic records while maintaining accuracy and consistency. At the heart of the process is OCR-based text extraction, which converts scanned documents into machine-readable text. Preprocessing techniques further enhance image quality, improving recognition rates and improving recognition rates and reducing errors [1], [4], [5], [9]. Beyond extraction, validation and classification modules ensure that the captured data is not only precise but also aligned with institutional requirements and standards [14], [17]. In this way, the system balances efficiency with reliability, offering a structured approach to academic document verification.

Guidelines for using the algorithms

The System is designed to follow established algorithmic procedures to ensure optimal performance in document verification. Text extraction is handled through OCR algorithms, specifically tesseract, while preprocessing techniques such as binarization and noise reduction are applied to enhance accuracy [1], [5], [9]. Once the text is extracted, a rule base validation system built on IF-THEN logic cross-checks the data against predefined institutional rules, ensuring compliance and consistency [14], [17]. Beyond text, classification and detection tasks are carried out using natural language processing (NLP) and convolutional neural



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networks(CNNs). These algorithms not only identify the type of document but also detect key visual features, including institutional Logos and seals[3], [11], [37], [38]. Together, these components create a layered verification process that balances efficiency with reliability, reducing errors while strengthening institutional trust in academic records.

User-Centric Design

The system is intentionally built around a user-centric design, making it straightforward for registrars and other staff to operate without unnecessary complexity. Tasks such as uploading documents or reviewing verification results require minimal effort, reducing the learning curve and improving overall efficiency. This design philosophy aligns with research emphasizing that usability is a crucial factor in a system adoption, particularly in administrative environments where time and accuracy are paramount [46], [47]. By prioritizing ease of use, the system not only enhances productivity but also encourages consistent engagement, ensuring that verification processes remain both reliable and crucial.

Testing Tools and procedure

The system's design follows a user-centric methodology, making it straightforward and efficient for registrars to operate. The goal is to ensure that the interface feels intuitive, requiring little effort to perform essential tasks such as uploading documents or reviewing verification results. By emphasizing usability, the system reduces barriers to adoption and supports smoother workflows. This approach reflects findings in prior research, which highlight that ease of use is a crucial factor in the successful implementation of administrative systems and in achieving operational efficiency [46], [47].



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Research Literature

Studies on automated document verification emphasize that preprocessing and OCR accuracy are critical to system performance. The techniques such as CLAHE(Contrast Limited Adaptive Histogram Equalization), Gaussian smoothing, and laplace sharpening significantly boost recognition rates, particularly for documents captured under poor lighting or with uneven contrast [5], [6], [7], [8], [9]. Comparisons between OCR engines reveal trade-offs: Tesseract performs well on clean, high resolution scans but struggles with noisy or distorted inputs[1],[3]. In more challenging scenarios, deep learning frameworks like EasyOCR built on CNNs and LSTMs achieve superior results, often surpassing 95% accuracy [2], [10], [11], [12]. Handwritten text recognition has also advanced through hybrid approaches that combine transformer-based models with conventional OCR techniques, further improving reliability [12], [13].

OCR Accuracy and Image Preprocessing

The quality of input images plays a decisive role in how well OCR systems perform. If the scanned or photographed document is noisy, low-contrast, or poorly lit, recognition accuracy drops significantly. That's why preprocessing techniques such as noise reduction, binarization, and contrast enhancement are often applied before OCR is run. Research shows that these steps can dramatically improve text recognition rates [5], [6]. More recent approaches even incorporate machine learning into preprocessing, refining the image in ways that adapt to specific document conditions [7]. Taken together, these developments highlight that



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preprocessing isn't just a helpful add-on; it's a critical stage in ensuring that data extracted from scanned documents is both reliable and accurate [8].

Document Verification Systems

Document verification systems are designed to confirm both the validity and accuracy of information contained in official records. In the academic setting, OCR-based verification has become increasingly valuable, allowing universities to process transcripts, enrollment forms, and other documents more efficiently [3], [4]. By automating validation, these systems reduce the risk of human error while speeding up administrative workflows. More advanced platforms now integrate OCR with artificial intelligence, enhancing reliability and precision. This combination is particularly effective in detecting inconsistencies or fraudulent documents, ensuring that academic records remain trustworthy and secure [41], [33].

Real-Time Document Processing and Automation

Real-time document processing makes it possible for systems to extract, analyze, and validate data the moment a document is received. This capability is especially valuable in educational institutions, where large volumes of records need to be managed quickly and accurately. Technologies such as GPU-accelerated OCR and edge computing have been shown to support real-time performance, boosting both speed and reliability [27], [28]. Studies also highlight that applying OCR in real-time settings improves the responsiveness and efficiency of academic workflows [22], [25]. When combined with automation, these systems further reduce manual intervention, streamline operations, and enhance overall productivity in registrar and administrative offices [46], [48].



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Review of Related Literature

OCR and Preprocessing Methods

OCR performance depends heavily on the quality of the input image, and studies consistently emphasize this point. When documents are scanned under poor lighting or with uneven contrast, recognition accuracy drops sharply. Preprocessing techniques such as CLAHE, Gaussian smoothing, and Laplace sharpening help counter these issues by redistributing pixel intensity, reducing noise, and sharpening edges. These adjustments make text clearer and allow OCR engines to extract information more reliably, even from low-quality scans. More recent developments go a step further, using machine learning–based preprocessing that dynamically adapts image parameters to the document’s condition. This evolution shows that preprocessing isn’t just a supportive step, it’s a foundational part of producing reliable OCR outputs.

Rule-Based Validation

Rule-based validation frameworks strengthen reliability by applying logical conditions derived directly from institutional regulations. Using deterministic “if–then” rules, they check critical fields such as names, dates, and course codes, ensuring that documents comply with established standards and remain transparent. One of their key advantages is interpretability; registrar staff can easily trace why a particular record fails validation, which builds trust in the process. At the same time, the literature points out an important limitation: these systems are not static. To remain effective, they must be continuously updated and aligned with evolving



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institutional policies, otherwise their checks risk becoming outdated or incomplete.

Deep Learning Approaches

Deep learning frameworks such as EasyOCR, which leverage CNNs and LSTMs, have proven to outperform traditional OCR engines like Tesseract when dealing with noisy, skewed, or otherwise imperfect images. CNNs excel at extracting spatial features from visual data, while LSTMs maintain sequential context, making the recognition process more robust in challenging conditions. This synergy allows EasyOCR to handle complex layouts and degraded inputs with higher accuracy.

Hybrid approaches are also gaining traction by combining transformer-based models with conventional OCR techniques; systems can adapt to diverse document types and significantly improve handwriting recognition. Transformers bring contextual awareness and long-range dependencies into the recognition process, complementing the strengths of CNNs and LSTMs. Together, these methods represent a shift toward more adaptive, intelligent verification systems that can handle the variability of real-world academic documents.

Feature Extraction and Classification

Beyond text recognition, modern verification systems combine OCR with natural language processing (NLP) to automatically categorize certificates for example, distinguishing between honors, workshop attendance, or course completion records. Reducing human intervention in this process not only speeds up workflows but also minimizes errors. Visual



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authentication methods add another layer of reliability: institutional markers such as logos and seals can be detected to confirm authenticity. Deep CNNs further strengthen verification by analyzing signatures, offering biometric security while also ensuring that documents comply with institutional formatting and design standards. Together, these techniques create a multi-layered verification framework that balances efficiency, accuracy, and trustworthiness in academic record management.

Limitations

OCR performance, despite its advances, still faces notable limitations tied to image characteristics and system requirements. Low-resolution or poorly scanned documents yield weak recognition results, underscoring the dependence of OCR accuracy on input quality. While deep learning-based models have reached near human-level accuracy, they demand substantial computational resources, which makes institutions with limited infrastructure hesitant to adopt them. Even multimodal verification methods struggle when confronted with highly irregular layouts or diverse handwriting styles, revealing gaps in adaptability. Moreover, formatting-based verification approaches are constrained by rigid design standards documents that deviate even slightly from institutional templates may be flagged incorrectly. These challenges highlight the trade-offs between accuracy, scalability, and practicality in deploying automated academic document verification systems.



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Review of Related Studies

Efficiency Gains in Registrar Offices

Practical implementations demonstrate that automation significantly enhances registrar workflows. Real-time feedback systems are particularly effective, as they immediately flag errors in academic documents, reducing delays and boosting productivity. Pilot projects in Philippine universities provide concrete evidence: automated transcript verification systems achieved faster turnaround times compared to traditional manual methods, streamlining the process for both staff and students. These findings reinforce the idea that automation not only improves efficiency but also strengthens institutional reliability in handling academic records.

Hybrid Models

Research shows that hybrid validation frameworks combining rule-based logic with machine learning are highly effective in automated document verification. Rule-based models provide deterministic checks (e.g., "if-then" conditions for names, dates, or codes) that are transparent and easy for registrar staff to interpret. Machine learning models, on the other hand, introduce adaptability, allowing the system to handle varied document formats and evolving institutional policies. When integrated, ensemble techniques balance these strengths: fixed rules ensure interpretability and compliance, while supervised learning models enhance flexibility and accuracy. This trade-off creates systems that are both reliable and adaptable, capable of evolving alongside institutional requirements without sacrificing transparency.



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Certificate Classification and Visual Verification

Automated classification systems streamline registrar workflows by extracting structured data such as student names, course titles, and issuing organizations directly from documents. This reduces manual effort in activity point allocations and ensures consistency. To strengthen authenticity checks, visual features are incorporated: logo detection and seal recognition commonly implemented through CNNs add institutional markers to the verification process, making it more reliable. Signature verification systems, also leveraging deep CNNs, have shown significant success in reducing forgery detection errors. Their effectiveness extends beyond academic contexts into legal domains, where biometric security and formatting compliance are critical. Together, these techniques create a layered verification framework that integrates textual, visual, and biometric dimensions, enhancing both accuracy and trustworthiness in document validation.

Administrative Efficiency

Automation has a profound impact on registrar workflows by cutting down the time spent on repetitive, low-value tasks. Studies show that by automating verification processes, registrar offices can save hundreds of work-hours annually. This efficiency gain allows staff to redirect their efforts toward higher-level administrative responsibilities, such as policy oversight, student engagement, and strategic planning. Real-time automation also reduces bottlenecks, ensuring that document validation is not only faster but more reliable, ultimately strengthening institutional trust and productivity.



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Challenges

Despite the clear benefits of automation, several disadvantages remain. Heavy reliance on automated systems can erode institutional memory and reduce the role of human judgment, particularly when staff become dependent on outputs without fully understanding the decision-making logic behind them. Resource requirements also pose a challenge: real-time OCR and AI-based verification demand powerful computing infrastructure, which may be beyond the reach of institutions with limited budgets. Another critical issue is interpretability. Many machine learning models function as “black boxes,” making it difficult for registrar staff to understand why a document is invalidated. This lack of transparency complicates adoption, as trust in the system depends not only on accuracy but also on the ability to explain decisions in a clear, human-readable way.

Theoretical Background

This study is grounded in several theoretical and computational models that provide the foundation for the proposed automated academic document verification system. Specifically, it draws upon **Optical Character Recognition (OCR) Theory**, which explains the principles of converting visual text into machine-readable data; the **Rule-Based System Theory**, which emphasizes deterministic validation through logical “if-then” conditions; and the **Intelligent Document Processing (IDP) Framework**, which integrates OCR with artificial intelligence to enable adaptive, scalable, and context-aware document analysis. Together, these theories highlight the diverse computational strategies for extracting, analyzing, and



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validating textual data, ensuring that the system is both technically robust and aligned with institutional requirements.

Optical Character Recognition (OCR) Theory

Optical Character Recognition (OCR) theory sits at the intersection of **pattern recognition** and **computer vision**, providing the computational foundation for converting visual text into machine-readable form. At its core, OCR systems rely on a pipeline of **image preprocessing, segmentation, and feature extraction**. Preprocessing enhances clarity (e.g., noise reduction, contrast adjustment), segmentation isolates individual characters or words, and feature extraction identifies structural patterns such as strokes or contours. These steps enable recognition of both printed and handwritten text from digitized documents.

Modern OCR systems extend this classical pipeline with **machine learning and deep learning models**. Convolutional Neural Networks (CNNs) excel at capturing spatial hierarchies in character shapes, while Recurrent Neural Networks (RNNs) preserve sequential dependencies, making recognition more accurate across varied text flows. Together, these models allow OCR to discern intricate character patterns even in distorted, noisy, or low-quality images. This evolution demonstrates how OCR theory has progressed from deterministic rule-based recognition toward adaptive, data-driven approaches that achieve near human-level accuracy.

In the context of your study, OCR functions as the **central mechanism** for transforming scanned academic documents such as transcripts and certificates into machine-readable text. This extracted text then becomes the **primary input** for the document verification process,



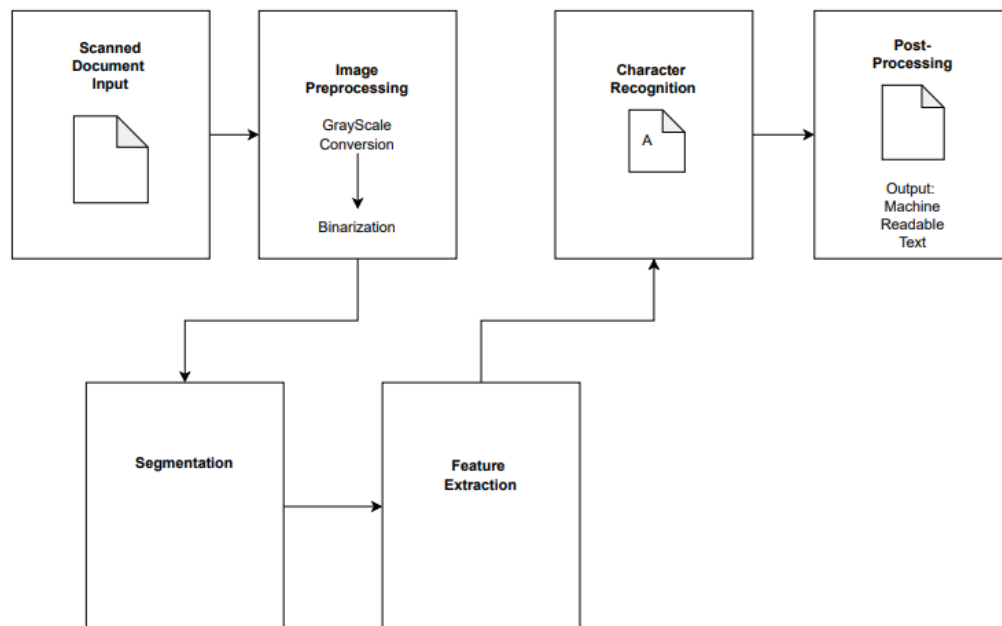
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enabling subsequent validation steps like rule-based checks, NLP categorization, and visual authentication. By anchoring the workflow on OCR, the system ensures that unstructured visual data is converted into structured information, which can be systematically analyzed for accuracy, authenticity, and compliance with institutional standards.

The OCR process used in this study is illustrated in Figure 1, which shows the transformation of scanned document images into machine-readable text through image preprocessing, feature extraction, and character recognition.



(Figure 1. Optical Character Recognition Process Model)



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Rule-Based System Theory

Rule-Based System Theory, rooted in artificial intelligence and expert systems, provides a structured approach to decision-making through **predetermined logical rules**. These systems represent knowledge using *IF-THEN* constructs, where conditions are evaluated to determine whether specific criteria are met. When the input satisfies the rule conditions, the system produces a corresponding output or decision, ensuring consistency and transparency.

In this study, **rule-based validation** functions as a **quality control mechanism** for information extracted from academic documents. It operationalizes institutional validation rules to verify the **presence, correctness, and formatting** of essential elements such as student names, identification numbers, course codes, and dates. This approach ensures that extracted data aligns with institutional standards before being accepted into the verification pipeline, thereby reinforcing accuracy and compliance.

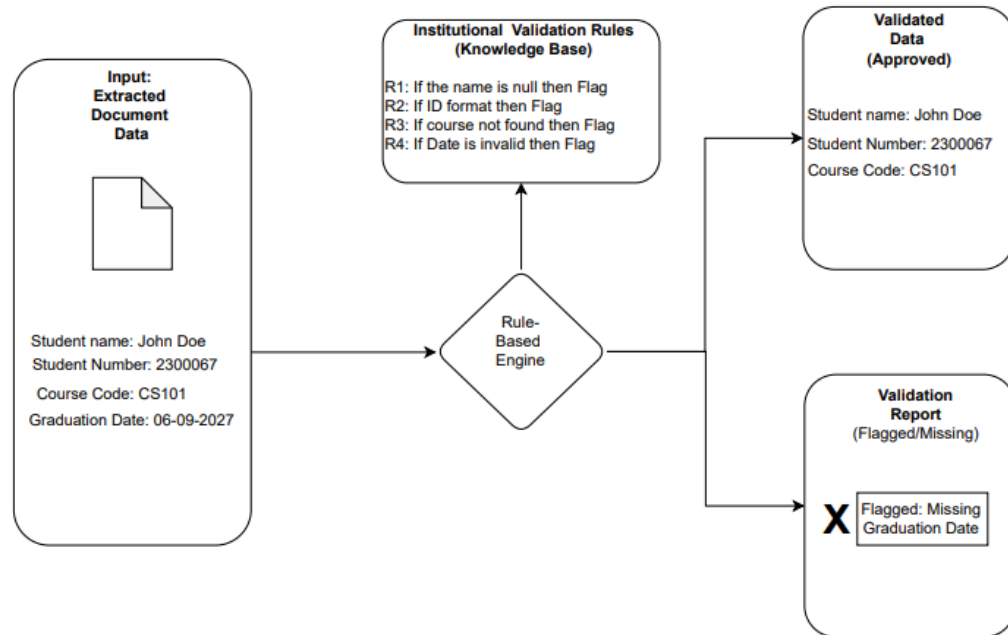
When the extracted information relies on violating any of the set rules, the system marks the document as incomplete or invalid. The rule-based validation approach used in this paper is depicted in Figure 2 which illustrates the procedure of evaluation of document data extracted against the set validation rules



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(Figure 2: Rule-Based Validation Model)

Conceptual Framework

The conceptual framework of the proposed Automated Real-Time Academic Document Verification System illustrates the relationship between the **input**, **process**, and **output** components that drive the system's functionality. Each element plays a critical role in ensuring that academic documents are verified efficiently, accurately, and in compliance with institutional policies.



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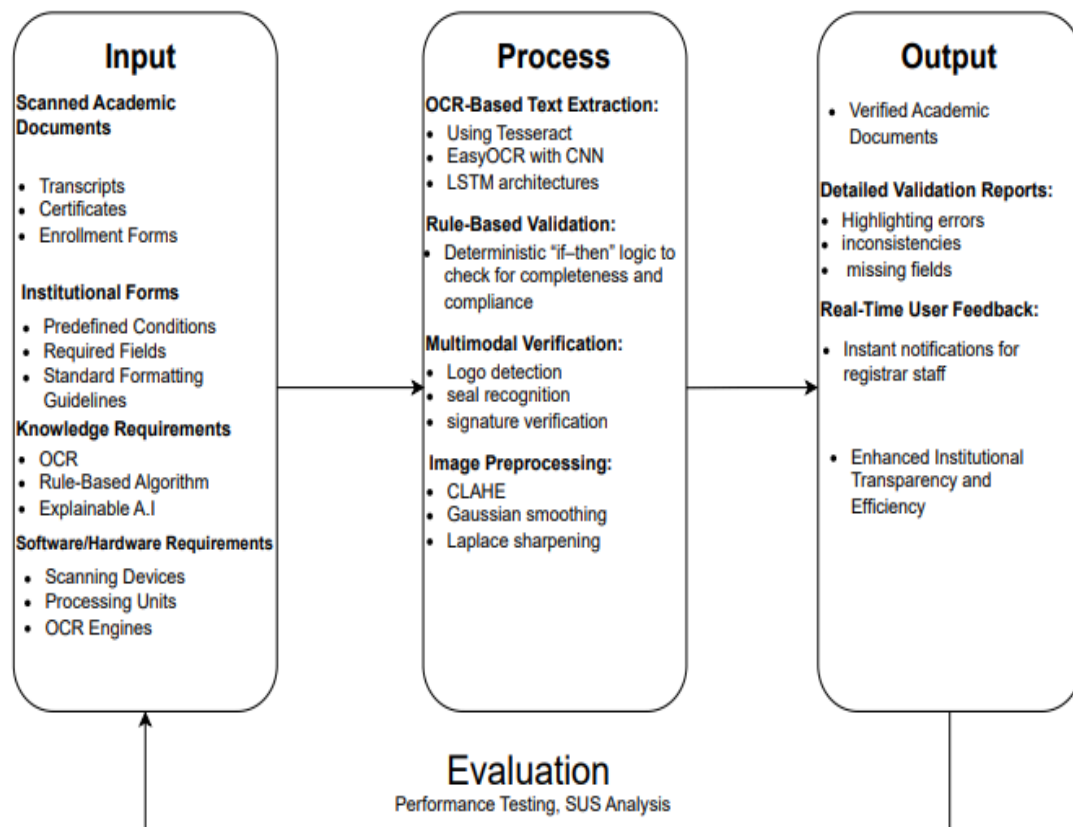


Figure 3: Input Process Output

The conceptual framework of our study is structured using the **Input–Process–Output (IPO) model**, which illustrates how academic document verification can be automated while maintaining compliance, accuracy, and transparency. This paradigm reflects the logical flow of the system, from data acquisition to algorithmic processing and final output delivery, and is firmly grounded in the objectives stated in Chapter I, the literature reviewed in Chapter II, and the methodology detailed in Chapter III.



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Input is where everything begins. The system first collects the essentials: scanned academic records like transcripts, certificates, and diplomas. These documents rarely look the same, different layouts, formatting quirks, and varying quality make preprocessing a must if we want reliable recognition. Alongside the raw files, institutional rules and formatting policies step in as the standards. They define what “complete,” “accurate,” and “compliant” really mean. For instance, registrar offices often require names, course codes, or grading structures to follow strict formats. Just as important are the needs of registrar staff, whose workflows and usability expectations shape how the system should function. By blending raw data, institutional criteria, and user requirements, the input stage ensures the system is firmly grounded in its real-world context.

Process is the heart of the system, the part where automation does the heavy lifting. It starts with OCR engines like Tesseract or EasyOCR, which convert scanned text into machine-readable data. To boost accuracy, preprocessing techniques such as CLAHE, Gaussian smoothing, and Laplace sharpening clean up the images before recognition. Then, deep learning models CNNs, LSTMs, and transformer-based hybrids step in to handle tricky cases like skewed scans or handwritten notes. Once the text is extracted, rule-based validation checks it against institutional policies using deterministic logic and Regular Expressions. Missing fields, wrong formats, or inconsistencies are flagged automatically. To go beyond text, multimodal verification adds layers like logo detection, seal recognition, and signature checks, helping guard against forgery or tampering. Together, these steps make the process phase a robust engine for trustworthy verification.



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Output is where everything comes together. The system produces verified documents that meet institutional standards, ready for official release. If something's off, it generates detailed validation reports so staff can quickly spot and fix issues. Real-time feedback also plays a role, alerting registrar staff immediately and cutting down delays. This feedback loop doesn't just improve efficiency it builds accountability and reduces manual effort. In short, the outputs aren't just technical results; they're practical tools that make registrar operations smoother and more reliable.

By organizing the system into input, process, and output, the IPO model shows how automation can be both technically sound and tailored to registrar workflows. Features like multimodal verification and explainable AI reinforce authenticity and transparency, ensuring trust in the results. Ultimately, this approach doesn't just digitize documents, it validates them with speed, precision, and accountability, contributing to the broader digital transformation of higher education.



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Synthesis

Over the years, academic document verification has evolved from slow, error-prone manual checks to advanced AI-powered systems that integrate multiple technologies for speed and accuracy. Preprocessing techniques such as CLAHE, Gaussian smoothing, and Laplace sharpening now enhance OCR performance, while deep learning frameworks like EasyOCR, CNNs, LSTMs, and transformer-based hybrids consistently outperform older engines such as Tesseract, particularly when handling noisy scans or handwritten text. Beyond text extraction, modern systems employ multimodal verification logo and seal recognition, signature checks, formatting validation, and certificate classification to strengthen authenticity and reduce human oversight errors. Researchers highlight the importance of administrative efficiency, privacy safeguards, and transparency, with explainable AI frameworks fostering trust and GPU acceleration enabling real-time processing. Challenges remain in image quality, computational demands, and policy adaptation, yet the synthesis underscores the need for registrar workflows, especially in Philippine higher education, to adopt systems that are both technically advanced and contextually tailored. The proposed Automated Real-Time Academic Document Verification System embodies this vision by combining OCR, rule-based validation, multimodal checks, and explainable AI to deliver accuracy, efficiency, and transparency, representing not just a technical solution but a step toward the broader digital transformation of academic services ensuring verification is faster, more reliable, and ultimately more trustworthy.



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Definition of Terms

In writing this study, certain keywords and concepts appear repeatedly both in the technical design of the system and in its practical use within registrar operations. To avoid confusion and ensure clarity, this section provides the **Definition of Terms**. The conceptual terms focus on the algorithms, methods, and frameworks that form the backbone of the system, while the operational terms explain how the system functions in everyday registrar tasks.

Placing these definitions at the start of the study serves an important purpose: it helps readers, whether they come from a technical background or an administrative one, share a common understanding of the language being used. This shared vocabulary not only makes the discussion easier to follow but also highlights how the system works in practice showing its potential as a tool for more reliable academic document verification in higher education institutions.

Conceptual Terms

Term	Definition
Certificate	Automated categorization of certificates using OCR and NLP [36].
Classification	



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Contrast Limited Adaptive Histogram Equalization(CLAHE)	Preprocessing technique that enhances local contrast in scanned images [5], [7]
EasyOCR	Deep learning-based OCR framework using CNN and LSTM [2], [10], [11], [12].
Explainable Artificial Intelligence (XAI)	Artificial intelligence techniques that provide interpretable outputs for transparency [20], [21].
Formatting Verification	Automated techniques that check compliance with formatting standards [42].
Information Systems Success Model (ISSM)	Framework for evaluating system quality, information quality, user satisfaction, and net benefits [8].
Legal Document Authentication	Systems that detect forgery using OCR, NLP, and CNNs [41].



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Logo Detection and Recognition

Image processing and deep learning methods for identifying institutional logos [37], [38].

Optical Character Recognition (OCR)

Technology that converts scanned or image-based text into machine-readable data [1], [3], [5], [11].

Rule-Based Validation

Logical verification process that evaluates extracted data according to predefined institutional rules [14], [15], [16].

Seal Recognition

Hybrid systems designed to recognize text in circular seals [39].

Signature Verification

Biometric process that verifies handwritten signatures using deep learning [40].

Tesseract OCR

Open-source OCR engine for text extraction [1], [3], [24], [25].



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Operational Terms

Term	Definition
Academic Document Verification	The process of evaluating academic records such as transcripts, certificates, and enrollment forms to ensure completeness, accuracy, and compliance using the proposed automated system.
Automated Verification System	The developed software application that performs document verification tasks automatically using OCR-based text extraction and rule-based validation.
Document Uploading	The procedure of submitting scanned or digital academic documents into the system by registrar staff for processing and verification.
Preprocessing Stage	The initial phase of document handling where image enhancement techniques are applied to improve clarity and prepare the document for accurate text extraction.
Text Extraction Process	The operation in which the system retrieves textual information from uploaded documents through OCR technology.



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Validation Process	The stage where extracted data is evaluated against predefined institutional rules to determine its correctness and completeness.
Validation Rules	A set of predefined institutional standards and conditions used by the system to check the accuracy, format, and completeness of extracted data.
Error Detection	The system's capability to identify missing, incorrect, or inconsistent information within academic documents.
Real-Time Feedback	Immediate notifications generated by the system to inform users about validation results, including detected errors and inconsistencies.
Verification Report	A system-generated output that summarizes the results of document validation, including identified errors and overall document status.
Document Status (Valid/Invalid)	The classification assigned by the system indicating whether a document meets all validation requirements or contains errors.
Multimodal Verification	The use of multiple verification methods, including text validation, logo detection, seal recognition, and signature verification, to ensure document authenticity.



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User Interface (UI)	The platform through which registrar staff interact with the system, including uploading documents and viewing verification results.
System Accuracy	The degree to which the system correctly verifies and validates academic documents compared to the original content or manual verification.
Processing Time	The amount of time required by the system to complete the verification process from document upload to result generation.
Usability	The level of ease with which registrar staff can learn, operate, and interact with the system effectively.
System Reliability	The consistency of the system in producing accurate and dependable verification results over repeated use.
Data Compliance	The extent to which academic documents meet institutional formatting and content requirements based on validation rules.
Registrar Workflow Integration	The ability of the system to align with and support the existing processes and operations of the registrar's office.



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User Acceptance

The willingness of registrar staff to adopt and use the system based on its perceived usefulness, usability, and reliability.

CHAPTER III

RESEARCH METHODOLOGY

This chapter discusses the methodology and procedures followed in developing and evaluating the Automated Real-Time Academic Document Verification System used. This part briefly presents the research design, locale, population and sample, and data-gathering procedures. This also discusses the algorithms used, the tools and instruments utilized, validation procedure, evaluation and scoring criteria, and ethical considerations observed in the study.

Research Design

The study uses Developmental Research Design, it is appropriate for developing and refining a technological enhancement or solution to a real-world problem. In this case, the problem to be solved is the traditional manual verification of academic documents, which is indeed time-consuming, prone to human error, and difficult to determine. This design allows the researchers to design and develop the system and continuously refine it through testing and feedback.

The System Development Life Cycle (SDLC) framework was adopted to systematize the development process. It presents a systematic order of processes, starting with the requirements analysis and moving on to system design, implementation, testing, and evaluation. Following the Software Development Life Cycle(SDLC), the researchers can be sure that every stage is executed systematically and, thus, there is less risk of neglecting key aspects and creating a more efficient and stable system.



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A quantitative research approach was employed to evaluate the system's performance, focusing on measurable aspects such as OCR accuracy, processing time, and user satisfaction ratings. These results were then compared with traditional manual verification processes to determine whether the proposed system provides a significant improvement. This approach is appropriate because it relies on collecting objective, quantifiable data that can be analyzed statistically.

Furthermore, the study adopts the agile development model to facilitate continuous improvement throughout the development process. Unlike traditional linear methodologies that follow a fixed sequence of steps, agile allows developers to repeatedly build, test, and refine the system. The iterative approach ensures that the system can be enhanced based on feedback and test results. For instance, if the OCR module performs inadequately on certain documents, adjustments can be implemented immediately in the subsequent development cycle rather than waiting until the final stages. Such flexibility is particularly valuable in practical settings, such as registrar offices, where a wide variety of documents must be processed efficiently and accurately.

Research Locale

The study will be carried out at University of Cabuyao(Pamantasan ng Cabuyao) Office of the University Registrar, which is located at Katapatan Mutual Homes Barangay Banay-Banay, City of Cabuyao, Laguna. We were selected as the research locale for the processing and verifying academic documents such as official school letterhead, school logo, authorized signature, dry seal & stamp, and document format. Conducting the study in this environment allowed the researchers to evaluate the proposed system under real operational conditions and ensure that it meets the practical needs of the office. Consequently, it serves as an ideal setting for testing the proposed system, as the institution stands to benefit from faster processing and greater accuracy in handling documents.



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Respondents of the Study

The respondents of this study consisted of 10 registrar staff who were selected through purposive sampling. They were chosen based on the direct involvement in academic document verification, as they serve as the primary users of the system. Their participation ensured that the evaluation reflected the actual workflow and challenges encountered during registrar operations.

In addition, 5 IT experts were included to provide a technical perspective. These professionals were tasked with evaluating the system's performance, usability, and reliability from a software engineering and systems development standpoint. Their involvement contributed to a balanced assessment by combining functional evaluation from end-users with technical aspects of the system thoroughly examined.

For the testing phase, a total of 40 academic documents were used as a dataset. This included 20 transcripts, 10 certificates of enrollment, and 10 diplomas, which represent the most commonly processed records in registrar operations. These materials were deliberately selected to reflect the typical documents handled by registrar staff on a daily basis. By utilizing actual and commonly used academic documents, the study ensured that the system was tested under realistic conditions.

The dataset enabled the evaluation of the system's OCR accuracy, the effectiveness of rule-based validation, and the reliability of multimodal verification across different document types. Furthermore, the inclusion of diverse document categories provided a comprehensive understanding of how the system performs in addressing the varying requirements of registrar tasks.



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Classification	Number of Respondents
Registrar Staff	10
IT Experts	5
Total:	15

Table 1: This table presents the analysis of the proposed system on the basis of usability, reliability, efficiency, and general usefulness to enhance the academic document verification process. Their feedback will give a significant idea about the performance of the system and how it could be used to improve the performance of the Registrar Office.

Data Gathering Procedures

The data gathering process was planned carefully to make sure the system was tested in a real-life setting while keeping academic records safe and private. The testing was done at the Registrar's Office of Pamantasan ng Cabuyao, where the usual tasks of the staff were copied or simulated.

Each document used in the study was first scanned and then processed by the system. The system cleaned the image, read the text using OCR, and checked the information based on set rules. The results were then checked to see if they were accurate, correct, and reliable. This helped the researchers know if the system can match or improve the manual way of checking documents.

After testing the system, the registrar staff were asked to use it themselves. They did tasks similar to their daily work, like checking transcripts, certificates, and diplomas. After that, they answered the System Usability Scale (SUS) survey to share their opinions about how easy, fast, and reliable the system is.



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To make the test fair, all participants were given the same documents and instructions. The researchers also observed the process and recorded the time, errors, and any problems they noticed. These were written in the Performance Testing Log.

This process helped test the system in a realistic situation while keeping personal information safe by removing real details and using sample data instead. By combining system testing and user feedback, the researchers were able to gather useful data to evaluate how effective the system is.

Instrumentation and Validation

The tools and validation process in this study were carefully planned to check how well the system works and how acceptable it is to users. Different tools were used to collect both numerical and descriptive data. A performance testing log was used to record how fast the system processes documents and how accurate the OCR is. The OCR accuracy was measured using the Character Error Rate (CER), which shows how well the system reads scanned text.

To check how well the system classifies documents, a confusion matrix was used. This helps determine if the documents are correctly labeled as Compliant or Non-Compliant. To measure how easy the system is to use, the researchers used the System Usability Scale (SUS), which is a 10-question survey given to the registrar staff. This helps evaluate the system's usability, reliability, and effectiveness in real situations.

The system was also tested and validated in several steps to make sure it is accurate and reliable. First, pilot testing was done using a small number of documents to find problems and improve the system. This helped fix errors early before full testing. Next, IT experts and registrar staff reviewed the system to check how well it works and how easy it is to use, and they gave suggestions for improvement. Lastly, statistical methods were used to check the reliability of the survey results. Cronbach's Alpha was used to see if the responses in the SUS survey were consistent. Through these steps, the researchers made sure that both the system's performance and user satisfaction were properly evaluated.



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Discussions on Algorithms/Mathematical Concepts Used/Proposed Solutions

The proposed system integrates a combination of algorithms and mathematical models to ensure accurate text extraction, reliable validation, and robust verification of academic documents. Each algorithm was carefully selected based on its suitability for handling the challenges identified in the literature, such as noisy scans, inconsistent formatting, and the need for authenticity checks.

The system OCR algorithm primarily uses EasyOCR, which is built on Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) models. CNNs are effective in recognizing visual patterns in scanned text, while LSTMs handle sequential dependencies, making them suitable for reading continuous text lines. EasyOCR was chosen because of its ability to process complex document layouts and noisy inputs. For benchmarking purposes, Tesseract OCR is also employed as a baseline. Tesseract is a widely used open-source OCR engine, but it performs best on clean, high-resolution documents. Comparing EasyOCR against Tesseract allows the researchers to highlight improvements in accuracy and adaptability.

The preprocessing algorithm used since OCR accuracy depends heavily on input quality, preprocessing is a critical step. The system uses CLAHE (Contrast Limited Adaptive Histogram Equalization). Enhances local contrast, making faint or unevenly lit text more readable. Gaussian Smoothing reduces image noise and blurring, which helps OCR engines distinguish text from background artifacts. Laplace Sharpening improves edge detection, ensuring that text boundaries are clearer for recognition. Together, these preprocessing techniques prepare scanned documents for optimal OCR performance, especially when dealing with older or low-quality

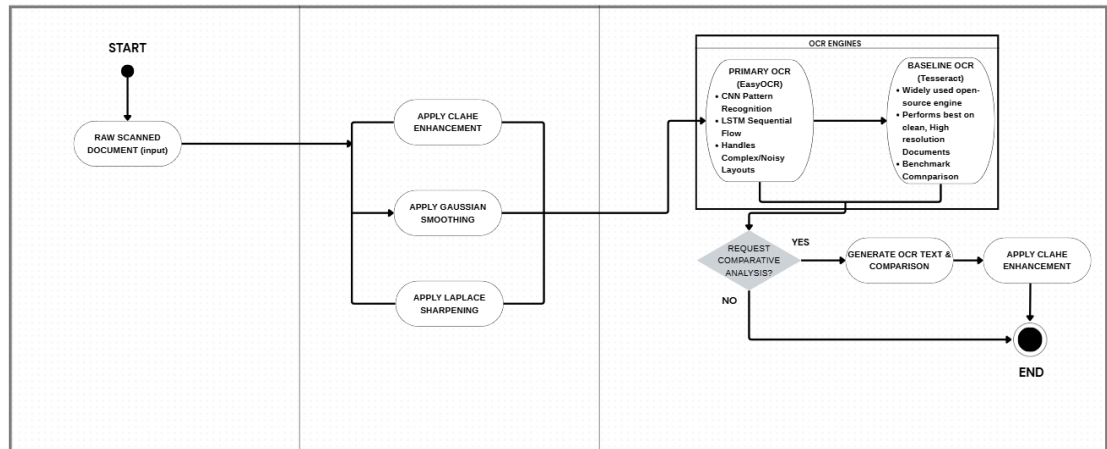


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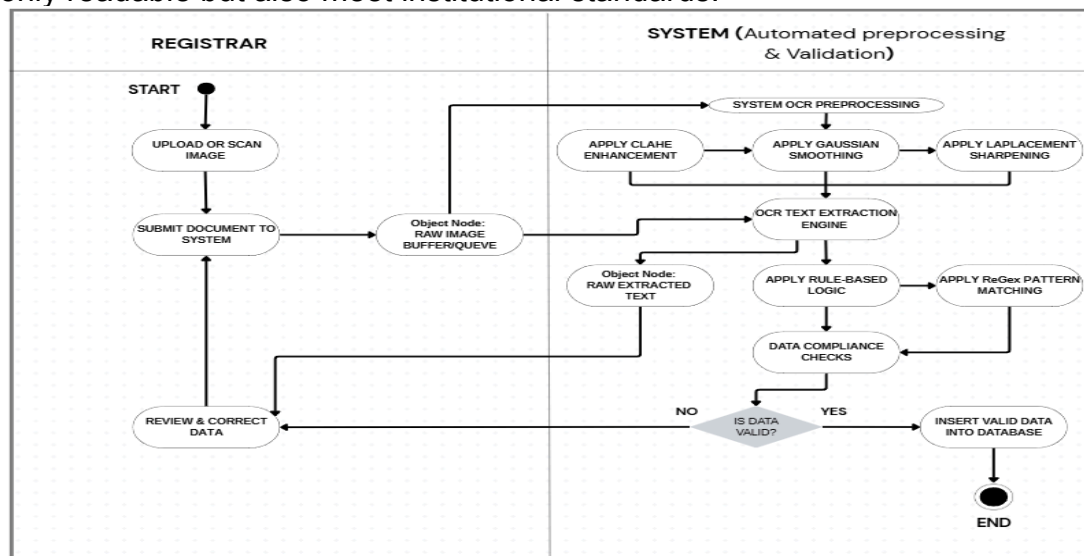
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scans.



The validation algorithms after text extraction, the system applies rule-based validation. This involves deterministic logic and Regular Expressions (Regex) to check for compliance with institutional policies. For example, Regex can detect whether dates follow the required format, whether names are present, or whether mandatory fields such as student ID numbers are filled. Rule-based validation ensures that documents are not only readable but also meet institutional standards.



The multimodal algorithm beyond text recognition, the system incorporates multimodal verification to strengthen authenticity checks the



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Logo Detection CNN-based models identify institutional logos, ensuring that documents originate from the correct source. Seal Recognition hybrid systems analyze circular seals, verifying both text and graphical patterns. Signature Verification deep learning models compare handwritten signatures against stored references, distinguishing genuine signatures from forgeries. These multimodal algorithms add an extra layer of security, addressing the risk of falsified or tampered documents.

System Development Methodology

The proposed project is built on the Agile Software Development Methodology, this method focuses on developing the system step by step as opposed to building it once. It is suitable for this study as it enables the system to be constructed gradually, including key components such as OCR feature, validation process, and user interface. Agile also supports regular testing and continuous user feedback, which helps to detect issues early and improve the system along the way. Through this process, the system can better meet the needs of the registrar staff and can be modified easily when necessary.



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Phases of Agile Software Methodology

System development follows the important phases of Agile software methodology:

1. Planning

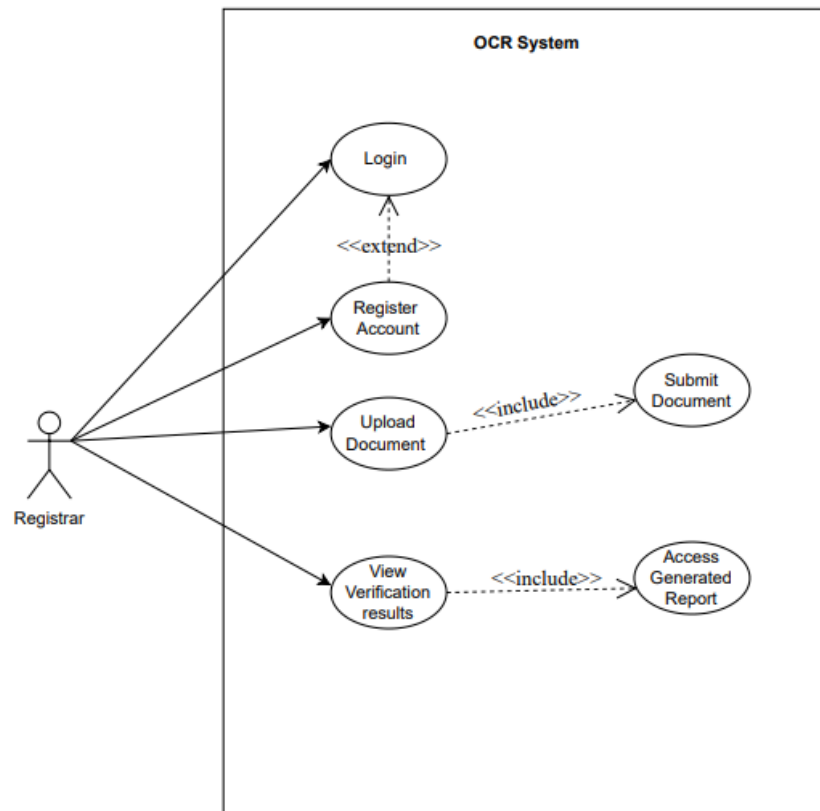


Figure 1: Use Case Diagram for planning phase

During the Planning phase, the researchers determined the major process of the workflow of manual document verification performed by the registrar office. In accordance with these observations, the system requirements were defined and represented using a use case diagram. The diagram shows the registrar as the primary actor who interacts with the



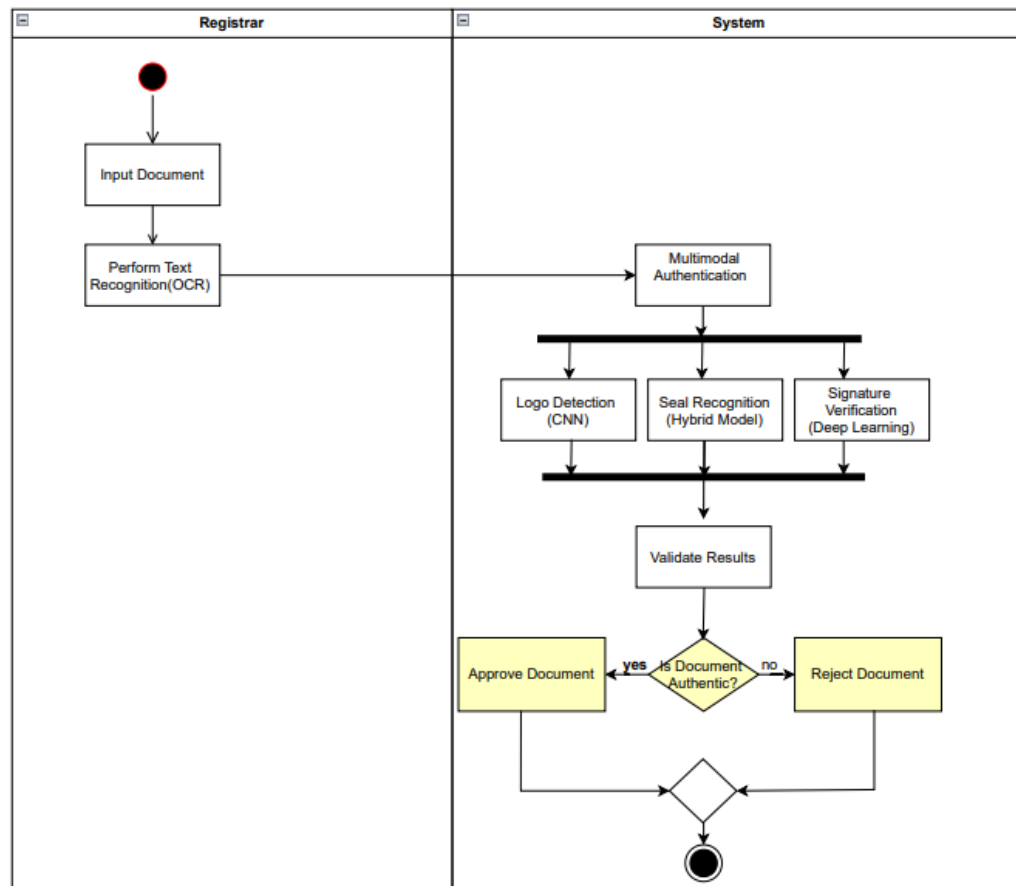
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system. The functions of the identified system are account management(Log in and register account), document processing(Upload Document and Submit document), and result evaluation(View verification results and access generated report). The relationship between register account and login, and the <<include>> relationships between upload document and submit document, as well as view verification results and access generated report. These interactions reflect the essential features needed to support an efficient and structured document verification process.

2. Design





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Figure 2: Activity Diagram For document authentication

The activity model illustrates the document authentication workflow where several verification strategies are employed. It shows the sequence of actions performed by both registrar and system. The process begins with the registrar uploading a scanned document into the system. After the upload, Optical Character Recognition(OCR) is applied to extract the textual data from the document. The workflow transitions to the system's backend, where multimodal authentication is performed. This phase involves three concurrent processes, namely logo detection with the help of a Convolutional Neural Network(CNN), seal recognition with hybrid model, and signature verification with the deep learning approach. The results of these processes are then consolidated and checked after they are done.

3. Development (Iteration)

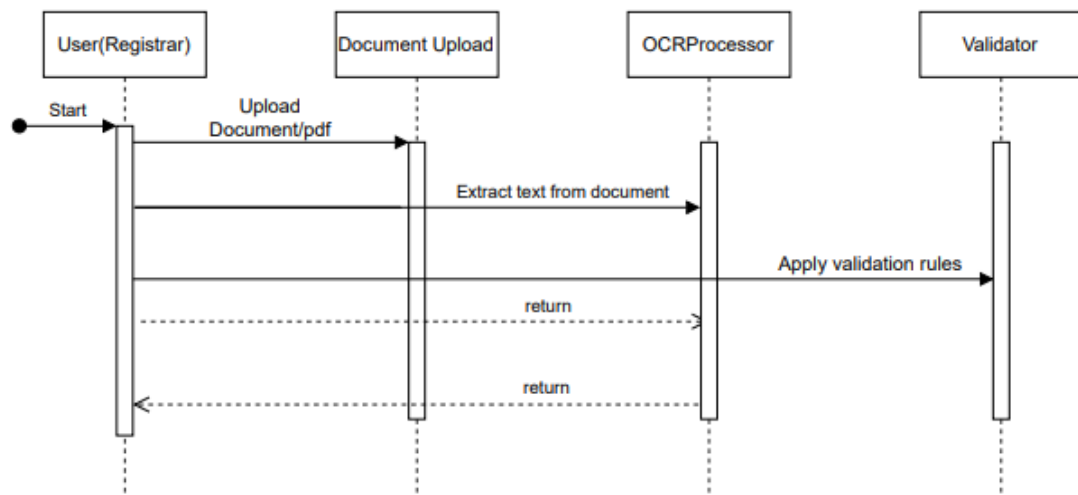


Figure 3: Sequence Diagram for Developing

During the Development phase, the system development was done in an iterative way and the various important parts of the system like uploading



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documents, OCR processing, validation logic and user interfaces were created and added to the system step-by-step. This approach ensures the efficient system development while allowing errors to be identified and corrected early. As a result, the overall accuracy and reliability of the system were continuously improved throughout the process.

4. Testing

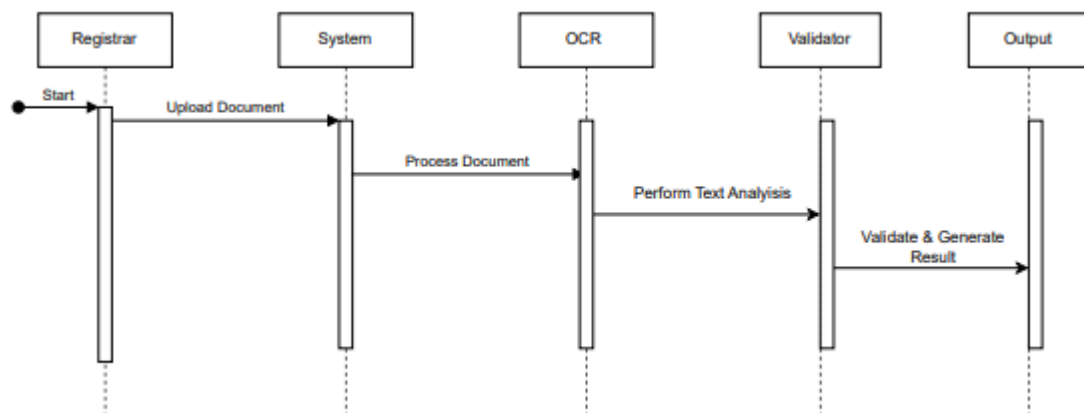


Figure 4: Sequence Diagram for testing phase

In Phase 4, a sequence diagram, as illustrated in figure 4, was utilized to represent the interaction and communication among system components during controlled test executions. The diagram helps visualize the step-by-step flow of processes, from the initial document upload by the registrar to the generation of the system output. Through this representation, the researchers were able to verify the sequential integrity of the document processing pipeline. It also ensured that key components, such as the OCR and validator modules, communicated correctly and system interactions were consistent with the intended architectural design.



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5. Deployment

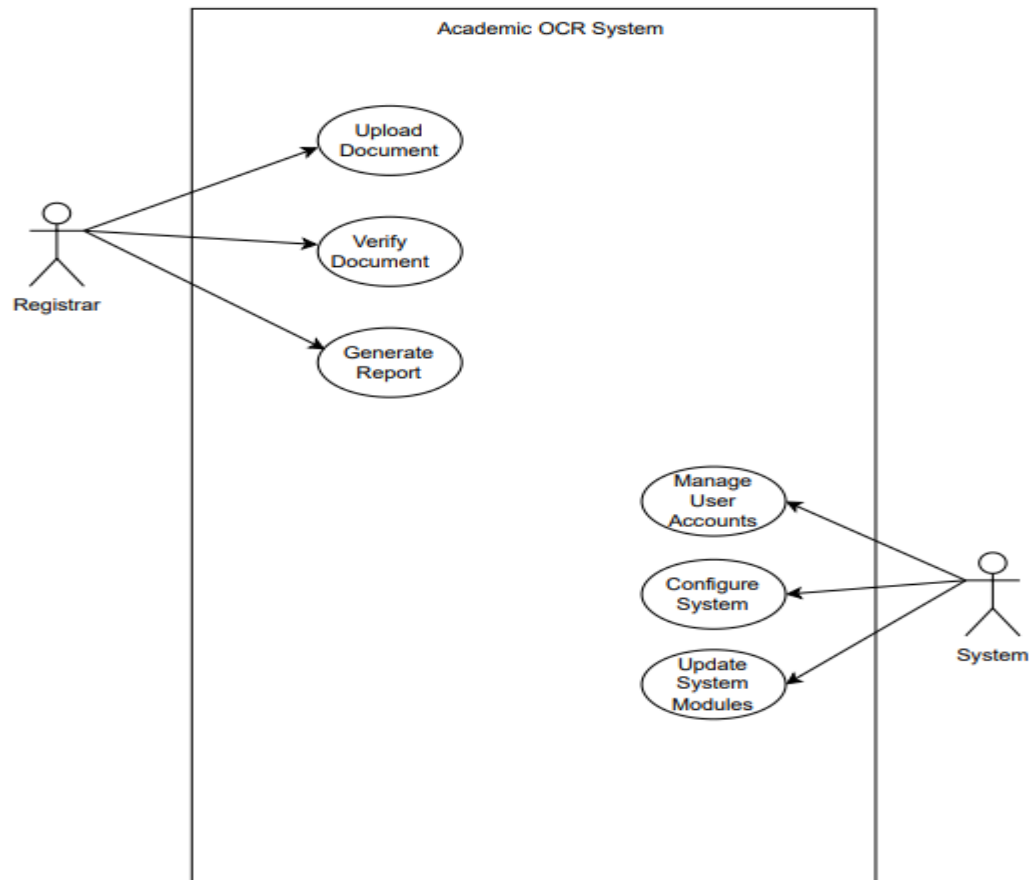


Figure 5: UseCase Diagram for Deployment

In figure 5, represents a use case diagram that illustrates the final functional scope of the deployed system. In this diagram, the two primary actors are identified: the registrar, who interacts with the system's front end features, and the system, which manages back end operations. The diagram clearly outlines the live functionalities available to the user, including uploading documents, verifying academic records, and generating managing



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user accounts and updating system modules, ensuring a comprehensive overview of all operational capabilities during deployment.

6. Review and Maintenance

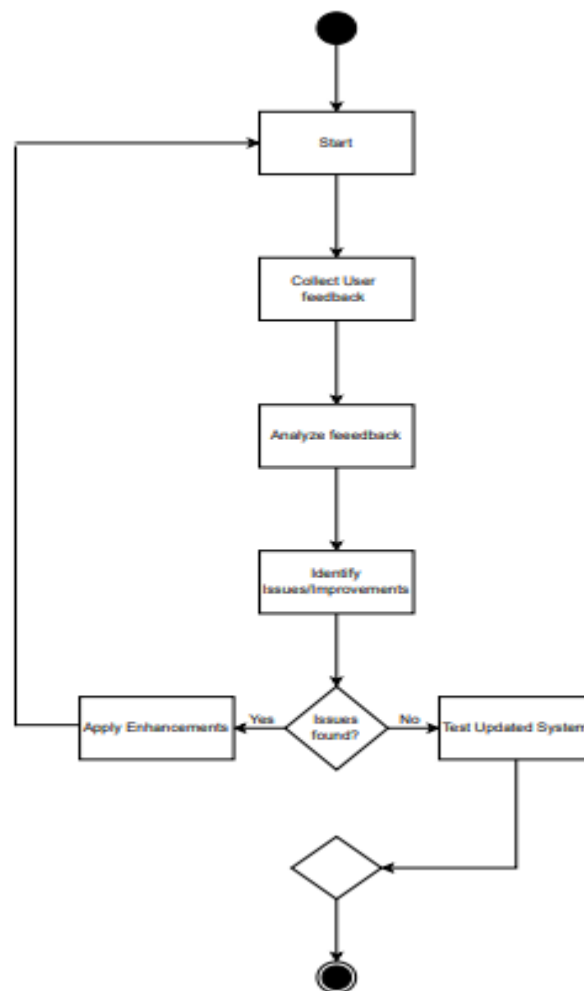


Figure 6: Activity Diagram for User Feedback

The review and maintenance phase represents the continuous improvement cycle within the agile methodology, ensuring that the system



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remains functional and relevant even after initial deployment. As illustrated in the activity diagram in figure 6, this phase begins with the collection and analysis of the feedback from the registrar staff to evaluate the software's performance. The feedback is examined through a structured sequence of activities, identifying specific issues or areas for enhancement, such as improving OCR accuracy for low quality documents or refining the validation interface.

A central decision point determines the subsequent course of the action. If issues or improvements are identified, the system undergoes an update cycle, during which modifications are implemented and rigorously tested to verify system stability before the release of the updated version. The iterative process demonstrates the system's capacity to evolve over time in response to user needs, thereby maintaining high levels of reliability and ensuring the integrity of the documents.

Statistical Treatment of Data

To accurately interpret the results and evaluate the performance of the proposed system, the following statistical tools and formulas will be utilized:

1. Weighted Mean

This will be used to analyze the responses from the Likert-scale survey (System Usability Scale) provided to the ten (10) Registrar Staff and five (5) IT Experts. It will determine the overall level of agreement regarding the system's usability, efficiency, and reliability.



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$$WM = \frac{\sum(f \cdot w)}{N}$$

Where: WM = Weighted Mean

f = Frequency of each response

w = Weight assigned to each response

N = Total number of respondents

2. Character Error Rate (CER)

To evaluate the technical performance of the OCR-based text extraction, the researchers will use CER. This measures the accuracy of the system by comparing the extracted text against the "ground truth" (the actual text on the document).

$$CER = \frac{S + D + I}{N}$$

Where: S = Number of Substitutions

D = Number of Deletions

I = Number of Insertions

N = Total number of characters in the reference

3. Confusion Matrix (Precision, Recall, and F1-Score)

This tool will be used to measure the effectiveness of the Rule-Based Validation module. It will track how many documents were correctly flagged as "Compliant" or "Non-Compliant."



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Precision: Measures the accuracy of "Non-Compliant" flags.

Recall: Measures the system's ability to find all actual errors in a document.

F1-Score: The harmonic mean of Precision and Recall, providing a single score for validation reliability.

4. Processing Time (Mean)

To validate the "Real-Time" aspect of the study, the arithmetic mean of the processing time (in seconds) will be calculated for the 40 test documents to ensure it falls within the targeted 1–2 second range.

Verbal Interpretation Scale

Range	Verbal Interpretation
4.21 – 5.00	Strongly Agree / Excellent
3.41 – 4.20	Agree / Very Good
2.61 – 3.40	Neutral / Good
1.81 – 2.60	Disagree / Fair
1.00 – 1.80	Strongly Disagree / Poor



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Population and Sampling

A few individuals took part here, their roles shaped by what the system could handle inside a classroom space. Documents appeared based on real tasks, pulled straight from daily work at Pamantasan ng Cabuyao's Registrar's Office. That spot came into play since clerks sort through stacks without digital help, each page demanding attention by hand. Time slips away there, energy drains fast under such weight. Testing unfolded where friction lives, right among paper trails piling up desk after desk.

From those involved in the project, fifteen people were chosen by the team. Among them, ten work in the registrar's office - handling routines that rely directly on the platform, giving firsthand insight into its day-to-day impact. Five others come from IT departments; their role brings a lens focused on stability, data safety, and precision.

Not just relying on people, the team pulled in 40 papers to run tests. Chosen carefully, these covered 20 Transcripts of Records, along with 10 Certificates of Enrollment and another 10 Diplomas. Different shapes and styles came into play here - each one pushing the system in its own way.

From chosen users came responses, then tested against those records gave solid outcomes. What emerged reveals performance clearly while pointing at gains possible within the registrar's workflow.

Evaluation and Scoring

Evaluation of the system relied on specific metrics to assess performance alongside usability. For reading precision, OCR accuracy formed the basis, with results expressed through Character Error Rate. Smaller values in CER indicate fewer mistakes during text capture. Document assessment capability received scrutiny via a confusion matrix instead. This



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method reveals effectiveness in distinguishing between Compliant and Non-Compliant cases clearly.

Checking system speed involved timing how long each document took to process. In contrast with standard manual review, performance differences became clear through comparison. Ease of use emerged via application of the System Usability Scale. Interpretation followed, categorizing outcomes as either Acceptable, Marginal, or Unacceptable.

With an aim to clarify findings, basic calculations came into play. From the SUS data, central tendency and spread emerged through average and variability measures. By applying a specific equation, precision of the system found its numerical expression

Accuracy Percentage From Document Verification

From this, the number of accurately handled files becomes clear. To examine whether the automated method takes notably less time than human review, a T-test was applied - offering insight into actual speed improvements. Though subtle, the contrast in duration matters.

Score	Interpretation	General Meaning
4	Excellent	Meets or exceeds expectations
3	Good	Acceptable with minor issues
2	Fair	Needs improvement
1	Poor	Does not meet expectations

Table : Score Interpretation.



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Criterion	Measurement Basis	4 (Excellent)	3 (Good)	2 (Fair)	1 (Poor)
OCR Accuracy	CER (%)	0–5%	6–15%	16–25%	>25%
Validation Accuracy	Accuracy (%)	90–100%	80–89%	70–79%	<70%
Efficiency	Time per document (sec)	1–2 sec	3–5 sec	6–10 sec	>10 sec
Usability	SUS Score	≥68	50–67	—	<50

Table : Criteria of Scoring



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Ethical Considerations

The conduct of this study strictly adhered to ethical standards to ensure the protection of participants, the confidentiality of data, and compliance with existing laws. In particular, the researchers observed the provisions of the **Data Privacy Act of 2012 (Republic Act 10173)**, which safeguards personal information from misuse and unauthorized disclosure.

All academic documents used during testing were carefully prepared to protect sensitive information. Identifiable details such as student names, addresses, and contact numbers were either redacted or replaced with dummy data. This ensured that no actual student records were exposed or permanently stored in the system. The dataset was created solely for research purposes and was handled with the highest level of confidentiality.

Participation of registrar staff was entirely voluntary. Before engaging in the study, respondents were informed of its objectives, procedures, and expected outcomes. They were assured that their feedback would be used only for academic purposes and that their identities would remain anonymous in all reports and publications. Informed consent was obtained, and participants were given the right to withdraw at any stage without penalty.

Confidentiality was maintained throughout the research process. Responses to the **System Usability Scale (SUS)** questionnaire were collected anonymously, and no identifying information was linked to individual feedback. Data logs and performance records were stored securely and accessible only to the research team.

Finally, the researchers committed to transparency and integrity in reporting results. Findings were presented honestly, without manipulation or misrepresentation, and limitations of the study were acknowledged. By observing these ethical considerations, the study ensured that both the participants and the institution were protected, while maintaining the credibility and trustworthiness of the research.



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Appendices

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CSA-3 AUTOMATED REAL-TIME ACADEMIC DOCUMENT VERIFICATION USING OCR-BASED TEXT EXTRACTION AND RULE-BASED VALIDATION (1).pdf

Turnitin

DOCUMENT DETAILS

Submission ID
2932

Submission Date
Apr 08, 2026 12:39 AM UTC

File Name
CSA-3 AUTOMATED REAL-TIME ACADEMIC DOCUMENT VERIFICATION USING OCR-BASED TEXT EXTRACTION AND RULE-BASED VALIDATION (1).pdf

File Size
928.4 KB

55
Pages

11,449
Words

113,388
Characters

*% detected as AI

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Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

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The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

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AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

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What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.





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EDUCATION

Bachelor of Science in Computer Science 2023-present
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- Currently studying

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School

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School

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WORK EXPERIENCE

POSITION

Company

- Indicate your accomplishments

Year

POSITION

Company

- Indicate your accomplishments

Year

LATEST SEMINAR ATTENDED

Association of Computer Science Student

The Great Flavor Byte
• February 27, 2026



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I am a third-year Computer Science student with a strong interest in software engineering, system design, and emerging technologies. My academic training has provided me with a solid foundation in programming, data structures, and algorithmic problem-solving, while also encouraging me to explore practical applications such as web development and database management. I am committed to continuous learning and aspire to contribute to innovative research and development in the field of computing.

EDUCATION

Bachelor of Science in Computer Science (BSCS) University of Cabuyao	2023-Present
Senior High University of Cabuyao • With Honors S.Y. 2022-2023 • Perfect Attendance	2021-2023
Secondary Alternative Learning System (ALS) • Best in Interview	2020-2021
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EDUCATION

Bachelor of Science Major in Computer Science (BSCS) 2023-
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- Currently studying

Pulo National High School S.Y 2020-2021
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School

- Graduated

WORK EXPERIENCE

POSITION

Company

Year

- Indicate your accomplishments

POSITION

Company

Year

- Indicate your accomplishments

LATEST SEMINAR ATTENDED

Organization

- JBECP SEMINAR WORK SHOP
- SEPTEMBER 2024

Title of Seminar

- Date Conducted

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EDUCATION

Bachelor of Science in Computer Science

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- Award Received With Honor

Elementary

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- Award Received None

WORK EXPERIENCE

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LATEST SEMINAR ATTENDED

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